Task 1 Report

Human Rule Formalization of §47 MBO
using the RASE Methodology

Course: Management von Informatikprojekten — RASE / Rule Formalization Project
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1. Introduction

1.1 Background

The construction industry relies heavily on legal and regulatory documents. These regulations are typically written for human interpretation, making them difficult to process automatically using computer systems.

To address this challenge, the **RASE methodology** provides a structured approach for converting legal text into machine-readable representations. RASE separates legal requirements into four categories:

Category	Meaning	Description
R	**Requirement**	The measurable rule that must be satisfied
A	**Applicability**	The entity or condition to which the rule applies
S	**Selection**	A specific sub-set or option within an applicability
E	**Exception**	A case or condition excluded from the rule

This structure enables legal requirements to be transformed into formal rules that can later be evaluated by software systems.

1.2 Project Objective

The objective of this project is to manually analyse **§47 of the Musterbauordnung (MBO)** using the RASE methodology and produce a correct human-generated reference model.

This reference model will later be used in **Task 2** to evaluate whether Large Language Models (LLMs) such as ChatGPT, Claude, Gemini, and Copilot can reproduce the same annotation automatically.

2. Selected Clause

2.1 Original Regulation — §47 Abs. 1 MBO

The following is the original German text of §47 Abs. 1 Musterbauordnung (MBO):

(Satz 1)	Aufenthaltsräume müssen eine lichte Raumhöhe von mindestens 2,40 m haben.
(Satz 2)	Aufenthaltsräume im Dachraum müssen eine lichte Raumhöhe von mindestens 2,20 m über mindestens der Hälfte ihrer Netto-Raumfläche haben; Raumteile mit einer lichten Raumhöhe bis zu 1,50 m bleiben außer Betracht.
(Satz 3)	Die Sätze 1 und 2 gelten nicht für Aufenthaltsräume in Wohngebäuden der Gebäudeklassen 1 und 2.

2.2 Why We Selected §47

This clause was selected because it contains all key RASE element types:

- Measurable requirements (specific height values in metres)
- Multiple applicability conditions (general rooms and attic rooms)
- Conditional and special rules (Dachraum exception)
- Legal exceptions (building classes 1 and 2)
- Measurement definitions (the 1.50 m area calculation rule)

This makes §47 an ideal candidate for RASE analysis and an appropriate test case for LLM evaluation.

2.3 Project History — From §36 to §47

The group initially worked on **§36 MBO**. After submitting this to Nick Nisbet for review, we received feedback that §36 was too large and too complex for the scope of this task. We therefore selected **§47-1 Aufenthaltsräume** as a more suitable clause.

3. Initial Legal Analysis

3.1 Sentence 1

> *"Aufenthaltsräume müssen eine lichte Raumhöhe von mindestens 2,40 m haben."*

Analysis:

- **Applicability (A1):** Aufenthaltsräume — the rule applies to all habitable rooms
- **Requirement (R1):** lichte Raumhöhe ≥ 2.40 m

3.2 Sentence 2

> *"Aufenthaltsräume im Dachraum müssen eine lichte Raumhöhe von mindestens 2,20 m über mindestens der Hälfte ihrer Netto-Raumfläche haben; Raumteile mit einer lichten Raumhöhe bis zu 1,50 m bleiben außer Betracht."*

Analysis:

- **Applicability (A2):** Aufenthaltsräume im Dachraum
- **Requirement (R2):** lichte Raumhöhe ≥ 2.20 m
- **Requirement (R3):** Over at least 50 % of net floor area
- **Note (NOT a RASE element):** "Raumteile mit einer lichten Raumhöhe bis zu 1,50 m bleiben außer Betracht." — This defines a measurement methodology, not a legal requirement (confirmed by Nick Nisbet — see Section 6).

3.3 Sentence 3

> *"Die Sätze 1 und 2 gelten nicht für Aufenthaltsräume in Wohngebäuden der Gebäudeklassen 1 und 2."*

Analysis:

- **Exception (E1):** Wohngebäude Gebäudeklasse 1 — Sätze 1 and 2 do not apply
- **Exception (E2):** Wohngebäude Gebäudeklasse 2 — Sätze 1 and 2 do not apply

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4. RASE Classification

The following table summarises the initial RASE element classification for §47:

Element	Type
Aufenthaltsräume	**Applicability**
Aufenthaltsräume im Dachraum	**Applicability**
lichte Raumhöhe ≥ 2.40 m	**Requirement**
lichte Raumhöhe ≥ 2.20 m	**Requirement**
≥ 50 % Netto-Raumfläche	**Requirement**
Wohngebäude GK1	**Exception**
Wohngebäude GK2	**Exception**

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5. Interpretation Challenges

This section documents the key semantic challenges encountered during the RASE annotation process.

Challenge 1 — Is Sentence 2 a Requirement or an Exception?

Initial interpretation: Sentence 2 initially appeared to be another independent requirement running in parallel with Sentence 1.

Resolved: After reviewing the clause structure and receiving expert feedback from Nick Nisbet, it became clear that Sentence 2 functions as a special rule for attic rooms (ExceptionSection) and therefore overrides the general rule from Sentence 1, rather than extending it.

Challenge 2 — The 1.50 m Statement — "bleiben außer Betracht"

Initial interpretation: The sentence "Raumteile mit einer lichten Raumhöhe bis zu 1,50 m bleiben außer Betracht." was initially interpreted as an Exception element and marked up accordingly.

Resolved: After receiving Nick Nisbet's final refinement e-mail (31 May 2026), it became clear that this sentence defines a method of measurement for net floor area — not a legal requirement on room design. It should therefore NOT be marked up as any RASE element at all.

6. Expert Feedback — Nick Nisbet (AEC3 / PhD Researcher)

Throughout the project we were in e-mail contact with **Nick Nisbet** (AEC3 UK Ltd, PhD researcher), an expert in RASE methodology and rule formalization. His feedback was essential in refining our final annotation.

Point 1: Aufenthaltsräume is itself an Application

The title "Aufenthaltsräume" at the beginning of the clause must itself be marked as an Applicability element. It cannot be treated as plain introductory text.

Point 2: Sentence 2 should be an ExceptionSection

Sentence 2 (the Dachraum rule) must be treated as a special ExceptionSection that overrides the general rule from Sentence 1. It is not an independent parallel requirement.

Point 3: The 1.50 m statement should NOT be marked up

The sentence about room parts with a clear height up to 1.50 m defines the method of measuring net floor area. It is not a requirement on the design and should receive no RASE mark-up at all. Nick's exact words: "It is part of a definition of the method of measurement, not a requirement on the design."

> *"By the way, it is very kind of you to say it, but I am not a professor, I am still writing my PhD!*
> *— Nick Nisbet, e-mail 29 May 2026"*

7. Final RASE Model

The following table presents the final validated RASE annotation of §47 Abs. 1 MBO, incorporating all expert feedback from Nick Nisbet.

ID	Type	Clause	Content	Metric / Property
A1	Applicability	Satz 1	**Aufenthaltsräume**	*type = Aufenthaltsraeume*
R1	Requirement	Satz 1	**Min. clear height 2.40 m**	*lichteRaumhoehe >= 2.40 m*
A2	Applicability	Satz 2	**Aufenthaltsräume im Dachraum**	*type = AufenthaltsraeumeImDachraum*
R2	Requirement	Satz 2	**Min. clear height 2.20 m**	*lichteRaumhoehe >= 2.20 m*
R3	Requirement	Satz 2	**Over ≥ 50 % of net floor area**	*nettoRaumflaeche >= 0.50 ratio*

ID	Type	Clause	Content	Metric / Property
E1	Exception	Satz 3	**Wohngebäude Gebäudeklasse 1**	*gebaeudeKlasse = GK1*
E2	Exception	Satz 3	**Wohngebäude Gebäudeklasse 2**	*gebaeudeKlasse = GK2*

Note: The sentence "Raumteile mit einer lichten Raumhöhe bis zu 1,50 m bleiben außer Betracht" (Satz 2) is intentionally excluded from the RASE model. Per expert feedback, it defines a measurement methodology, not a legal requirement.

8. Machine-Readable Interpretation

The final RASE model was transformed into IF/THEN logic rules suitable for computational interpretation. These rules serve as the machine-readable representation of §47 Abs. 1 MBO.

Rule R1

```
IF
Room = Aufenthaltsraum
AND Room != Dachraum
AND BuildingClass NOT IN {1, 2}
THEN
Height >= 2.40 m
```

Rule R2

```
IF
Room = Aufenthaltsraum
AND Room = Dachraum
AND BuildingClass NOT IN {1, 2}
THEN
AreaWithHeight >= 2.20 m
AND >= 50 % of NetArea
```

9. Human vs Machine Interpretation

In addition to the manual RASE annotation, AI-assisted analysis was conducted using Large Language Models (LLMs). The generated outputs were compared against the manually produced RASE model.

The comparison showed that AI can correctly identify many RASE elements. However, AI systems may struggle with semantic distinctions such as:

- Distinguishing measurement definitions from legal requirements (e.g. the 1.50 m statement)
- Recognising ExceptionSections that override general rules rather than extend them
- Correctly assigning Applicability scope across nested sentences

This highlights the importance of human validation in legal rule formalization and motivates the structured evaluation planned for Task 2.

10. Reflection

This project provided practical experience in legal rule formalization and demonstrated the challenges of translating natural language regulations into structured representations.

The most important lesson learned was that **legal interpretation requires understanding the meaning and context** of a clause rather than relying solely on keywords or surface-level text patterns. The 1.50 m measurement statement is a clear example: it reads like an exception but is actually a definitional rule about area calculation.

The collaboration with Nick Nisbet was invaluable — it confirmed that even experienced annotators need multiple rounds of review to produce a correct RASE model. The final annotation combines structured methodology with expert interpretation.

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11. Preparation for Task 2

The final annotation produced in Task 1 serves as the **ground-truth reference** for the next phase of the project.

Task 2 will investigate whether LLMs (ChatGPT, Claude, Gemini, Copilot) can reproduce the same RASE annotation automatically. Nick Nisbet outlined two possible approaches:

- **Option a:** Provide the LLM with examples of RASE mark-up created by colleagues (few-shot prompting)
- **Option b:** Write a detailed prompt with explicit RASE instructions and HTML tag specifications (instruction-based prompting)

The quality of the AI-generated annotations will be systematically compared against the human-generated reference model developed in this report, using a minimum of three prompt refinement iterations.